Heat-engine efficiency

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate useful output and rejected energy for a heat engine.

SUCCESS CRITERIA - I can...

1. Use efficiency=work output/heat input.
2. Account for rejected heat.
3. Distinguish an ideal upper bound from actual performance.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “efficiency” and “cycle” to explain your model or result. Sentence starters are optional.

efficiency: useful output divided by input

cycle: a process that returns a system to its initial state

kelvin: the absolute temperature scale used in thermal ratios

01 Bring back an earlier idea

At the same kelvin temperature, sample B contains 3 times as many particles of the same ideal gas as A.
Compare average energy per particle and total translational energy.

Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\eta = W_{out}/Q_{hot}$; $Q_{hot} = W_{out} + Q_{cold}$
$\eta_{max} = 1 - T_{cold}/T_{hot}$.
Cyclic heat engine; heat amounts are positive magnitudes. The reversible upper bound uses kelvin temperatures.

Worked reasoning: A heat engine takes 100 J from a hot source, does 25 J of work and rejects 75 J.
Efficiency=25/100=25%.

Your turn: Draw three labeled energy arrows and check conservation.

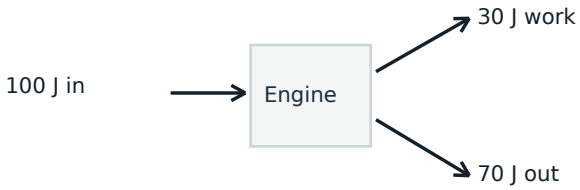
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

An engine receives 100 J, produces 30 J of work and rejects 70 J.
Check the energy balance and find efficiency.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Build the solution

A heat engine receives 200 J and produces 50 J useful work.
Find efficiency and rejected heat.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Heat-engine efficiency

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Why should a 95% efficient electric motor not be confused with a 95% heat engine converting heat from one source?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

At fixed mass, speed becomes 1/4 of its original value.
Compare kinetic energy and the energy-equivalent TNT amount.

Word help: kinetic energy = energy associated with motion

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

A device outputs 40 J work from 100 J heat.
State efficiency and rejected heat.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

An engine claims 100 J input, 40 J useful work and 50 J rejected heat.
Identify the missing accounting and give one possible explanation.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3